

Artificial Intelligence (AI) – Unit 1 Notes

1. Artificial Intelligence (AI)

1.1 What is AI?

Artificial Intelligence is the branch of computer science that focuses on building systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, perception, language understanding, and decision-making.

Formal definition: AI is the study and design of intelligent agents that perceive their environment and take actions to maximize the chances of achieving defined goals.

1.2 Goals of AI

- Automate intelligent behavior
 - Mimic human cognitive abilities
 - Enhance decision-making using data
 - Reduce human effort and error
 - Enable systems to learn and adapt
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1.3 Types of AI (Based on Capability)

1.3.1 Narrow AI (Weak AI)

- Designed for a specific task
- Cannot perform outside its domain
- Most AI systems today fall in this category

Examples: - Google Search - Netflix recommendations - Face recognition systems - Chatbots

1.3.2 General AI (Strong AI)

- Can perform any intellectual task a human can
- Possesses reasoning, learning, and adaptability across domains
- Still theoretical and not yet achieved

1.3.3 Super AI

- Surpasses human intelligence
- Self-aware and autonomous
- Hypothetical and part of future AI research

1.4 Types of AI (Based on Functionality)

1. **Reactive Machines** – No memory, reacts only to current input
 2. **Limited Memory** – Uses past data (most ML systems)
 3. **Theory of Mind** – Understands emotions and beliefs (research stage)
 4. **Self-aware AI** – Conscious systems (hypothetical)
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1.5 AI Use Cases

Healthcare - Disease diagnosis - Medical imaging analysis - Drug discovery

Finance - Fraud detection - Algorithmic trading - Credit scoring

Retail & E-commerce - Product recommendations - Demand forecasting - Chatbots for customer support

Manufacturing - Predictive maintenance - Quality inspection using vision

Education - Personalized learning systems - Automated grading

2. Machine Learning (ML) – Basic Overview

2.1 What is Machine Learning?

Machine Learning is a subset of AI that enables systems to learn patterns from data and improve performance over time without being explicitly programmed.

Key idea: Data → Patterns → Predictions

2.2 Types of Machine Learning

2.2.1 Supervised Learning

- Data is labeled
- Model learns input-output mapping

Algorithms: - Linear Regression - Logistic Regression - Decision Trees - Random Forest - Support Vector Machines (SVM)

Use cases: - Spam detection - Price prediction - Disease classification

2.2.2 Unsupervised Learning

- Data is unlabeled

- Finds hidden patterns

Algorithms: - K-Means Clustering - Hierarchical Clustering - DBSCAN - PCA

Use cases: - Customer segmentation - Market basket analysis

2.2.3 Semi-Supervised Learning

- Combination of labeled and unlabeled data
- Useful when labeled data is expensive

2.2.4 Reinforcement Learning

- Agent learns via rewards and penalties
- Based on trial and error

Use cases: - Robotics - Game playing (AlphaGo) - Traffic signal optimization

2.3 ML Workflow

1. Data collection

1. Primary Data Sources:

Primary data refers to data that is collected directly by an organization for a specific purpose or research objective. Examples: surveys/forms, sensors/IOT devices, Internal databases (SQL/ NoSQL)

2. Secondary data sources:

Secondary data refers to data that has already been collected by another organization and is reused for analysis. Examples: APIs, Web Scrapping, Pubic Datasets, external databases.

2. Data preprocessing

1. Cleaning and transforming raw data into a usable format for ML models.

2. It improves model accuracy and stability.

3. Steps:

1. Handling missing values (mean, median, drop)
2. Removing duplicates
3. Outlier detection
4. Encoding categorical variables (Label Encoding, One-Hot Encoding)
5. Feature scaling (Normalization, Standardization)

3. Feature engineering

1. Creating new meaningful features from existing data to improve model performance.

2. It aims to extract maximum predictive power from data.

3. Techniques:

1. Feature creation (Age from DOB)

2. Feature transformation (log transform)
 3. Interaction features
 4. Polynomial features
 5. Dimensionality reduction (PCA)
 6. Feature selection (RFE, Lasso)
 4. Model selection
 1. Choosing the most appropriate ML algorithm for the problem.
 2. It is decided based on problem type, dataset size, interpretability requirement, computational resources
 5. Training
 1. Process of feeding data to the model so it learns patterns.
 2. Goal is to Minimize loss function and improve generalization.
 3. Steps to be followed:
 1. Split data
 2. Fit model on training data
 3. Tune hyperparameters
 6. Evaluation
 1. Assessing how well the model performs on unseen data.
 2. Metrics:
 1. Regression:
 1. MAE
 2. MSE
 3. RMSE
 4. R^2
 2. Classification:
 1. Accuracy
 2. Precision
 3. Recall
 4. F1-score
 5. ROC-AUC
 7. Deployment
 1. Making the trained model available for real-world use.
 2. Methods:
 1. REST APIs (Flask, FastAPI)
 2. Cloud platforms
 3. Docker containers
 4. Mobile or Web apps
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3. Deep Learning (DL)

3.1 Introduction to Deep Learning

Deep Learning is a subset of Machine Learning that uses multi-layered neural networks to model complex patterns in large datasets.

Why Deep Learning? - Handles unstructured data (images, text, audio) - Learns features automatically - High accuracy for complex tasks

3.2 Difference: ML vs DL

Aspect	Machine Learning	Deep Learning
Feature extraction	Manual	Automatic
Data requirement	Low to medium	Very high
Computational power	Moderate	High (GPUs)
Interpretability	Higher	Lower

4. Neural Networks

4.1 What is a Neural Network?

A neural network is inspired by the human brain and consists of interconnected neurons (nodes) organized in layers.

4.2 Components

- Input layer
- Hidden layers
- Output layer
- Weights
- Bias
- Activation functions

4.3 Common Activation Functions

- Sigmoid
- Tanh
- ReLU
- Softmax

4.4 Types of Neural Networks

- Artificial Neural Network (ANN)

- Convolutional Neural Network (CNN) – images
 - Recurrent Neural Network (RNN) – sequences
 - LSTM / GRU – long-term dependencies
 - Transformers – attention-based models
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5. Natural Language Processing (NLP)

5.1 What is NLP?

NLP is a field of AI that focuses on enabling machines to understand, interpret, and generate human language.

5.2 NLP Processing Pipeline

1. **Text Acquisition**
 - Documents, emails, social media, reviews
 2. **Text Preprocessing**
 - Lowercasing
 - Tokenization
 - Stopword removal
 - Stemming / Lemmatization
 3. **Text Representation (Feature Extraction)**
 - Bag of Words (BoW)
 - TF-IDF
 - Word Embeddings (Word2Vec, GloVe, FastText)
 - Transformer embeddings (BERT)
 4. **Model Building**
 - Traditional ML models
 - Deep learning models
 - Transformers
 5. **Evaluation & Deployment**
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5.3 Core NLP Tasks

- Text classification
- Sentiment analysis
- Named Entity Recognition (NER)
- Part-of-Speech tagging
- Machine translation

- Question answering
 - Summarization
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5.4 NLP Use Cases

Business - Customer feedback analysis - Chatbots and virtual assistants

Healthcare - Clinical text analysis - Medical record summarization

Finance - News sentiment analysis - Risk assessment

Education - Automated essay grading - Plagiarism detection

6. Cognitive Computing

6.1 What is Cognitive Computing?

Cognitive computing refers to systems that simulate human thought processes such as understanding, reasoning, learning, and interaction.

It focuses on **augmenting human intelligence**, not replacing it.

6.2 Key Characteristics

- Context-aware
 - Adaptive and learning-based
 - Natural interaction (language, speech)
 - Probabilistic reasoning
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6.3 Cognitive Computing vs AI

Aspect	AI	Cognitive Computing
Goal	Automation	Human augmentation
Decision style	Deterministic	Probabilistic
Learning	Model-driven	Experience-driven
Interaction	Limited	Natural and conversational

6.4 Cognitive Computing Use Cases

- Clinical decision support systems

- Intelligent virtual assistants
 - Fraud investigation support
 - Personalized recommendations
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7. Relationship Between Concepts

AI → ML → Deep Learning → Neural Networks

AI also includes NLP, Cognitive Computing, Expert Systems, Robotics, and Computer Vision.

8. Summary

- AI is the umbrella concept
- ML enables learning from data
- DL uses neural networks for complex tasks
- NLP deals with human language
- Cognitive computing focuses on human-like reasoning and assistance

These technologies together power modern intelligent systems across industries.